Credit Default Prediction & Alternative Credit Scoring

Machine Learning case study for Credit Risk Modelling roles

CV PORTFOLIO

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PROJECT OVERVIEW

Alternative credit score model to predict loan default risk

Built a credit default prediction framework using LendingClub loan data. The project compares a traditional Logistic Regression benchmark with machine learning models and tests whether sentiment extracted from borrower narratives can add soft information to structured credit variables.

CLEAN SAMPLE
84,946
loan observations after preprocessing

DEFAULT RATE
16.9%
imbalanced classification target

MODELS
4
LR, Random Forest, SVM, XGBoost

BEST MODEL
XGBoost
highest AUC, Recall and F1

Business problem

Traditional credit scorecards can miss non-linear borrower risk patterns and may underperform for thin-file borrowers. The objective was to test whether machine learning and borrower narrative sentiment can improve default risk detection.

Headline result

XGBoost achieved the strongest overall credit-risk detection trade-off with AUC 0.7137 and Recall 0.6417. Random Forest had the highest accuracy, but its low recall made it less suitable for identifying actual default cases.

Relevant skills demonstrated

- Credit risk modelling
- PD / default prediction
- Feature engineering
- Class imbalance
- Text sentiment / NLP
- Model evaluation
- AUC / Recall / F1
- Risk interpretation

Methodology & Modelling Pipeline

How the project was structured as a credit risk modelling exercise

1. **Raw loan data** — LendingClub
2. **Clean & label** — Default vs non-default
3. **Engineer features** — Hard + soft data
4. **Train models** — LR, RF, SVM, XGB
5. **Evaluate risk** — AUC, Recall, F1

Feature design

Loan characteristics
loan amount, term, interest rate

Creditworthiness
delinquencies, recent inquiries, public records, total accounts

Solvency
annual income, DTI, revolving utilization, verification status

Alternative data
sentiment score from borrower description, title and purpose

Model setup

- Traditional benchmark: Logistic Regression to estimate default probability and provide an interpretable baseline.
- Machine learning challengers: Random Forest, Linear SVM and XGBoost to capture non-linear relationships and feature interactions.
- Text-to-feature approach: borrower narratives were cleaned and converted into sentiment scores using TextBlob.
- Imbalance handling: stratified split, random oversampling on training data and class-weighted or scale_pos_weight treatment in selected models.
- Evaluation focus: Accuracy, Precision, Recall, F1-score and AUC, with emphasis on Recall and AUC for default identification.

Credit risk logic translated into the project

- Default was treated as the target event, consistent with PD modelling logic.
- Only pre-approval variables were retained to reduce data leakage risk.
- Model comparison was interpreted through a risk lens: missing true defaults is costly, so Recall and AUC matter more than raw Accuracy.
- The final discussion included explainability, model selection, threshold tuning and risk governance considerations.

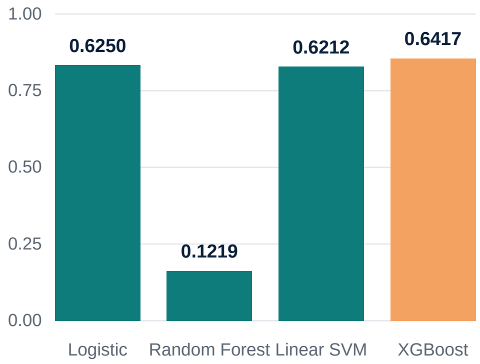
Model Performance & Risk Interpretation

Why XGBoost is the strongest candidate model for default risk detection

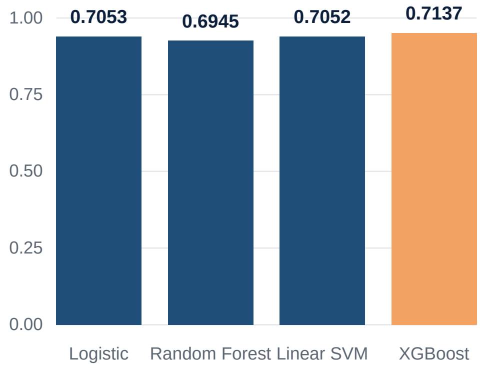
Performance comparison

Model	Accuracy	Precision	Recall	F1	AUC
Logistic	0.6686	0.2827	0.6250	0.3893	0.7053
Random Forest	0.8282	0.4692	0.1219	0.1935	0.6945
Linear SVM	0.6703	0.2833	0.6212	0.3891	0.7052
XGBoost	0.6613	0.2806	0.6417	0.3905	0.7137

Recall - ability to identify actual defaults



AUC - discrimination power



Key interpretation

XGBoost as best risk-detection

It achieved the highest Recall, F1-score and AUC. This makes it the strongest candidate when the business priority is to detect potential default borrowers rather than maximize overall accuracy.

Accuracy alone is misleading

Random Forest achieved the highest Accuracy, but its Recall was only 0.1219. In credit risk, this means the model missed many actual default cases, which would be dangerous for lending decisions.

Sentiment adds conceptual value

Borrower narratives can represent soft information. The current TextBlob sentiment variable was simple, but it demonstrated how unstructured text can be converted into a quantitative risk signal.

Practical credit risk takeaway

A deployable credit risk model should balance predictive power, default recall, interpretability, threshold strategy, monitoring and governance - not simply choose the model with highest accuracy.

CV-Ready Portfolio Summary

How to present the project for Credit Risk Model / Risk Analytics applications

Capability map for target roles

Credit risk modelling requirement	Evidence from this project
Default / PD modelling logic	Defined loan default target and built probability-based classification models.
Feature engineering	Combined loan terms, borrower creditworthiness, solvency indicators and sentiment feature.
Model development	Benchmarked Logistic Regression, Random Forest, Linear SVM and XGBoost.
Model validation mindset	Compared Accuracy, Precision, Recall, F1-score and AUC under imbalanced data.
Risk interpretation	Explained why Recall and AUC are more relevant than Accuracy for default detection.
Governance awareness	Identified model limitations, explainability issues and need for threshold tuning and monitoring.

CV-ready project bullets

- Built a credit default prediction framework using LendingClub data, combining structured borrower variables with sentiment-based alternative data from loan narratives.
- Benchmarked Logistic Regression, Random Forest, Linear SVM and XGBoost using Accuracy, Precision, Recall, F1-score and AUC; XGBoost achieved the strongest overall performance with AUC 0.7137 and Recall 0.6417.
- Applied credit risk modelling logic including default labelling, feature engineering, class imbalance handling, model comparison and business interpretation of risk-detection performance.

Interview talking points

1) Why Recall matters for default prediction.
2) Why Random Forest accuracy can be misleading under imbalance.
3) Why XGBoost captured non-linear borrower risk patterns better.
4) How sentiment represents soft information for alternative credit scoring.

Next improvements

Add SHAP explainability, ROC/PR curves, threshold optimization, PSI/model monitoring, PD calibration, reject inference discussion and domain-specific NLP methods instead of simple lexicon sentiment.

Recommended title to place on CV

Credit Risk Modelling Project - Alternative Credit Scoring using Machine Learning